

QUANTUM LDPC CODES

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SINGLE QUBITS AND BASIC ERRORS

- Unlike classical bits, a qubit exists as a superposition of computational basis states (the Z-basis):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- The X-Basis: Essential for understanding phase parity:

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

- The environment introduces physical errors:
 - Bit-Flip (Pauli X): Swaps $|0\rangle$ and $|1\rangle$. Detected via Z-basis measurement.
 - Phase-Flip (Pauli Z): Flips the relative phase. Detected via X-basis measurement.

THE FOUNDATIONS OF QUANTUM OPERATIONS

- The Building Blocks: Quantum errors and operations are defined by Pauli matrices, which all square to the identity

$$X^2 = I$$

$$Y^2 = I$$

$$Z^2 = I$$

- Multiplication Properties:

$$1. XZ = -iY$$

$$2. ZX = -XZ$$

- An n-qubit state is represented as a $2^n \times 1$ vector
- Tensor Products: Used to scale operations across multiple qubits

$$(A_1 \otimes A_2)(B_1 \otimes B_2) = (A_1 B_1 \otimes A_2 B_2)$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

INTRODUCTION TO STABILIZERS

- Measuring a qubit directly destroys its fragile quantum state.
- To solve this we group pauli operators into a “Stabilizer” (e.g., $M = X \otimes Z \otimes Z \otimes X$).
- Instead of measuring each qubit, a stabilizer measures the joint parity of the group.
- We encode our logical data $|\Psi\rangle$ so that it is perfectly immune to these stabilizers. It is defined mathematically as a +1 eigenstate:

$$M|\Psi\rangle = +|\Psi\rangle$$

- If a physical error occurs that anti-commutes with our stabilizer, it forces a negative sign.

TRANSLATING QUANTUM TO CLASSICAL

- Pauli operators are mapped into a classical binary array:

$(x \mid z)$

$$1.X \rightarrow (1 \mid 0) \quad 2.Y \rightarrow (1 \mid 1) \quad 3.Z \rightarrow (0 \mid 1) \quad 4.I \rightarrow (0 \mid 0)$$

- To check if an error and a stabilizer commute, we calculate their symplectic inner product classically:

$$\langle x_1, z_2 \rangle + \langle z_1, x_2 \rangle \pmod{2}$$

- The Result:
 - If 0: They commute (no error detected).
 - If 1: They anti-commute (the syndrome alarm triggers).

DEFINING THE QUANTUM PARITY CHECKS

- A stabilizer code space C is defined by a set of generator matrices: $C = \{ |\Psi\rangle \mid M_i |\Psi\rangle = |\Psi\rangle \}$
- Generators must pairwise commute, requiring our parity-check matrices to satisfy: **$H_x H_z^T + H_z H_x^T = 0 \pmod{2}$**
- Linear Independence: Generators must be independent (cannot be generated by multiplying others together).

Example Generators (5-Qubit System):

- $M_1 = XZZXI$
- $M_2 = IZZXX$
- $M_3 = XIXXZ$
- $M_4 = ZXIXZ$

EXTRACTING THE SYNDROME(s)

- A physical error E acts on the state, represented as $E=(e_x \mid e_z)$.
- The Parity Check Matrix: Our stabilizer generators form the rows of our global check matrix: $H=[H_x \mid H_z]$
- The Global Syndrome (s): The complete syndrome vector is calculated via block matrix multiplication modulo 2: $s=(H_x e_z^T + H_z e_x^T) \pmod{2}$
- For a single stabilizer i , the alarm triggers if the inner product is odd:
 $s_i = \langle h_{xi}, e_z \rangle + \langle h_{zi}, e_x \rangle$

PHYSICAL SYNDROME EXTRACTION

- Direct measurement of data qubits disturbs the encoded logical information. To prevent this, we extract the parity indirectly using an auxiliary "ancilla" qubit.
- The ancilla is prepared in a superposition state: $|0\rangle_A \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle_A + |1\rangle_A)$
- Controlled-Pauli (CM) gates are applied between the ancilla and the data qubits to map the parity information.
- A Hadamard (H) gate is applied to the ancilla to interfere the state.
- The ancilla is measured, yielding our classical syndrome bit $s_i \in \{0,1\}$.

STATE COLLAPSE EQUATION

- The Pre-Measurement State: Right before measuring the ancilla, our system is in an entangled superposition:

$$|\Psi_{\text{final}}\rangle = |0\rangle_A \otimes \left(\frac{I + M}{2}\right) |\psi\rangle_D + |1\rangle_A \otimes \left(\frac{I - M}{2}\right) |\psi\rangle_D$$

- The Projection (Collapse):
 - If $s_i=0$ (Commutes): The state collapses to $\left(\frac{I + M}{2}\right)$. Since $M = I$, the state evaluates strictly to $I|\psi\rangle_D$
 - If $s_i=1$ (Anti-Commutes): The state collapses to $\left(\frac{I - M}{2}\right)$. Since $M = -I$, the negatives cancel, evaluating to $I|\psi\rangle_D$
- The parity is measured, but the logical data remains perfectly intact.

WHAT DO WE DO WITH THE SYNDROME?

- After measurement of all ancilla qubits we end up with a classical binary vector s , where each bit indicates which stabilizer were violated.
- Now we need to find the most likely error that caused this syndrome pattern.
- It is a classical decoding problem where parity check matrix is binary and s also. We find e such that $He = s \pmod{2}$
- The quantum error correction reduces to classical decoding. The syndrome is our input and estimated error vector e is our output.

BP DECODER — CLASSICAL (Theoretical Baseline)

STEP 1: TANNER GRAPH CONSTRUCTION & CHANNEL INITIALIZATION

- Variable Nodes (VNs): Represent the received noisy bits or message elements.
- Check Nodes (CNs): Enforce parity-check constraints (the rows of H).
- Channel crossover prob: p
- Prior reliability: $L_0 = \log((1-p)/p)$
- If bit $c_i = 1$: $\text{LLR}[i] = -L_0$
- If bit $c_i = 0$: $\text{LLR}[i] = +L_0$

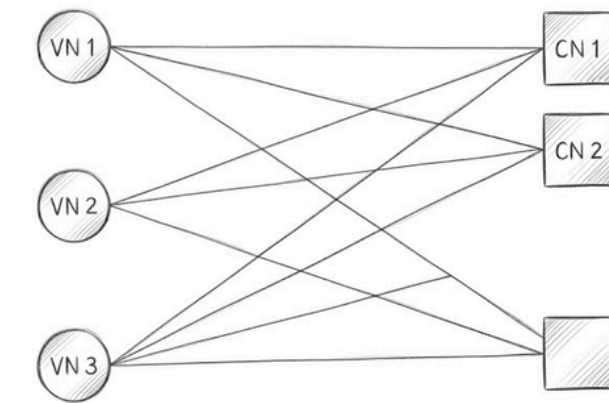
STEP 2: ITERATIVE MESSAGE PASSING (CHECK → VARIABLE → DECISION)

Variable → Check

$$m_{v \rightarrow c}^{(t)} = L_v + \sum_{c' \in N(v) \setminus c} m_{c' \rightarrow v}^{(t-1)}$$

Check → Variable

$$m_{c \rightarrow v}^{(t)} = 2 \tanh^{-1} \left(\prod_{v' \in N(c) \setminus v} \tanh \left(\frac{m_{v' \rightarrow c}^{(t-1)}}{2} \right) \right)$$



$$H\mathbf{c}^T = \mathbf{0}$$

DECISION & CONVERGENCE CHECK

- $\text{decoded}[i] = 1$ if total LLR < 0 else 0
- Check: $H \cdot \text{decoded} \equiv 0 \pmod{2} \rightarrow$ valid codeword
- else repeat until max iteration $\rightarrow$ best guess

BP DECODER — SYNDROME

STEP 1: ERROR MAPPING & SYNDROME-BASED INITIALIZATION

- Recovering the error vector $He = s \pmod{2}$
- Variable Nodes: qubit errors
- Check Nodes: Stabilizer measurements
- Channel crossover prob: p
- Prior reliability: $L_0 = \log((1-p)/p)$
- LLR initialize: All set to $+L_0$ (assume no error)

STEP 2: ITERATIVE SYNDROME INVERSION & CONVERGENCE

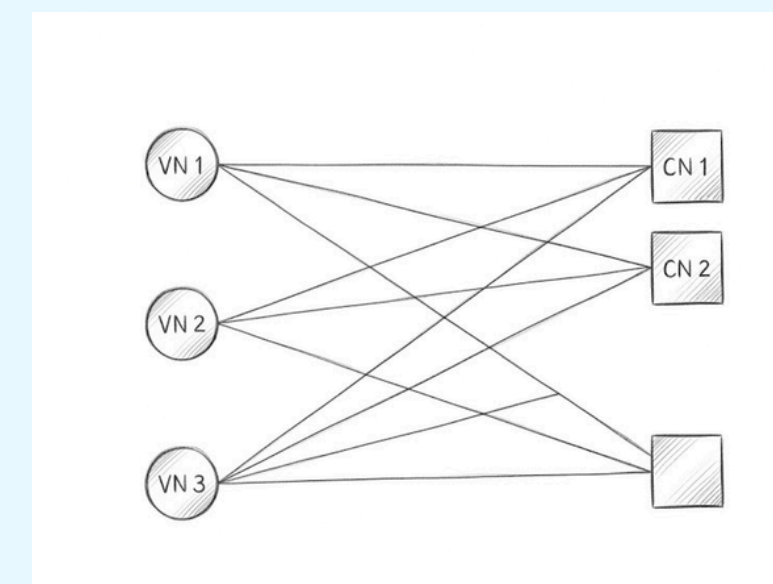
Variable $\rightarrow$ Check

$$m_{v \rightarrow c}^{(t)} = L_v + \sum_{c' \in N(v) \setminus c} m_{c' \rightarrow v}^{(t)}$$

Check $\rightarrow$ Variable

$$m_{c \rightarrow v}^{(t)} = 2 \tanh^{-1} \left(\prod_{v' \in N(c) \setminus v} \tanh \left(\frac{m_{v' \rightarrow c}^{(t)}}{2} \right) \right)$$

- Flip sign of product if $s[i]=1$



$$He = s \pmod{2}$$

DECISION & CONVERGENCE CHECK

- Check: $He = s \pmod{2} \rightarrow$ valid error vector
- else repeat until max iteration $\rightarrow$ best guess

WHY BP FAILS ON QUANTUM CODES

- Short Cycles in the Tanner graph.
- Degenerate Errors - multiple errors produce identical syndrome , BP can't distinguish them.

SOLUTION - OSD-0

- Sorting the H and e based on magnitude of LLR.
- Computing residual syndrome.
- Build information set.
- Solve $H_J x = s_{res}$
- Do gaussian elimination and undo permutation to get error vector

OSD-0 - FROM SOFT GUESS TO HARD CORRECTIONS

SORT BY RELIABILITY



```
blf_per = argsort(|belief|)
H_per = H[:, blf_per]
e_per = e[blf_per]
```

- Low |LLR| = unreliable, sorted last

RESIDUAL SYNDROME



```
s_res = s  $\oplus$  (H_per · e_per mod 2)
```

- What BP left unsolved

BUILD BASIS (RANK CHECK)



```
if rank([H_j | s_res]) == rank(H_j):
    break
if rank([H_j | col]) > rank(H_j):
    J.append(i); H_j = [H_j | col]
```

- independent set

SOLVE & APPLY CORRECTION

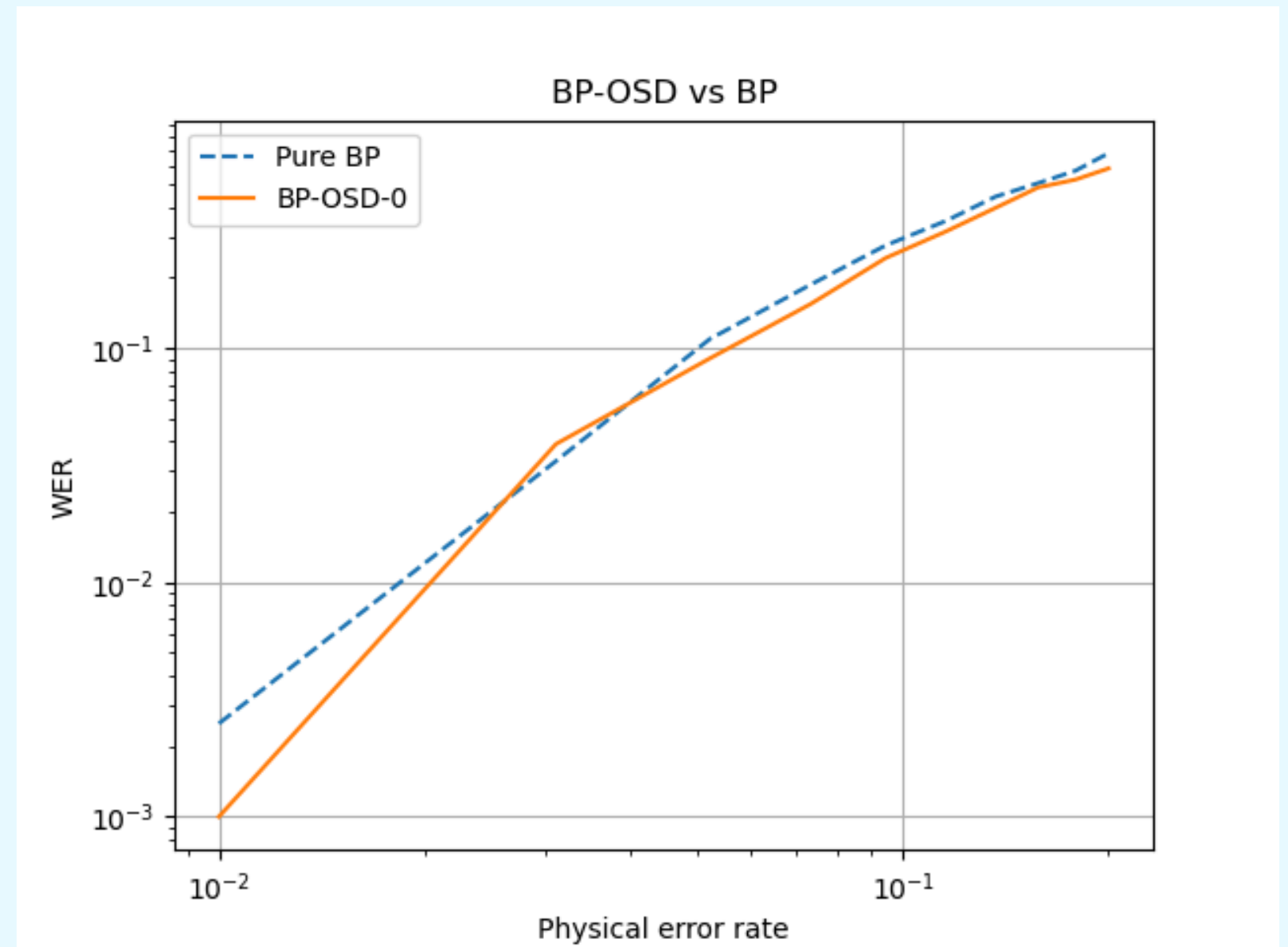


```
x = solve(H_j, s_res) # over  $F_2$ 
for k in range(len(J)):
    e_f_per[J[k]] = x[k]
e_f = e_f_per[r_blf_per]
```

- Map back to original qubit indices

CURVE

- The plot compares the Word Error Rate(WER) of two decoders with physical error rate.
- Physical error rate indicates how noisy the channel is and Word Error Rate indicates how often decoder fails to find the correct error.
- OSD post processing made improvement over BP alone. The gap indicates the error that BP could not resolve but OSD successfully resolved using reliably sorted Gaussian elimination.



RESULTS AND CONCLUSION

- **BP alone -**

Fail on short cycle and degenerate errors in quantum LDPC codes

- **BP + OSD -**

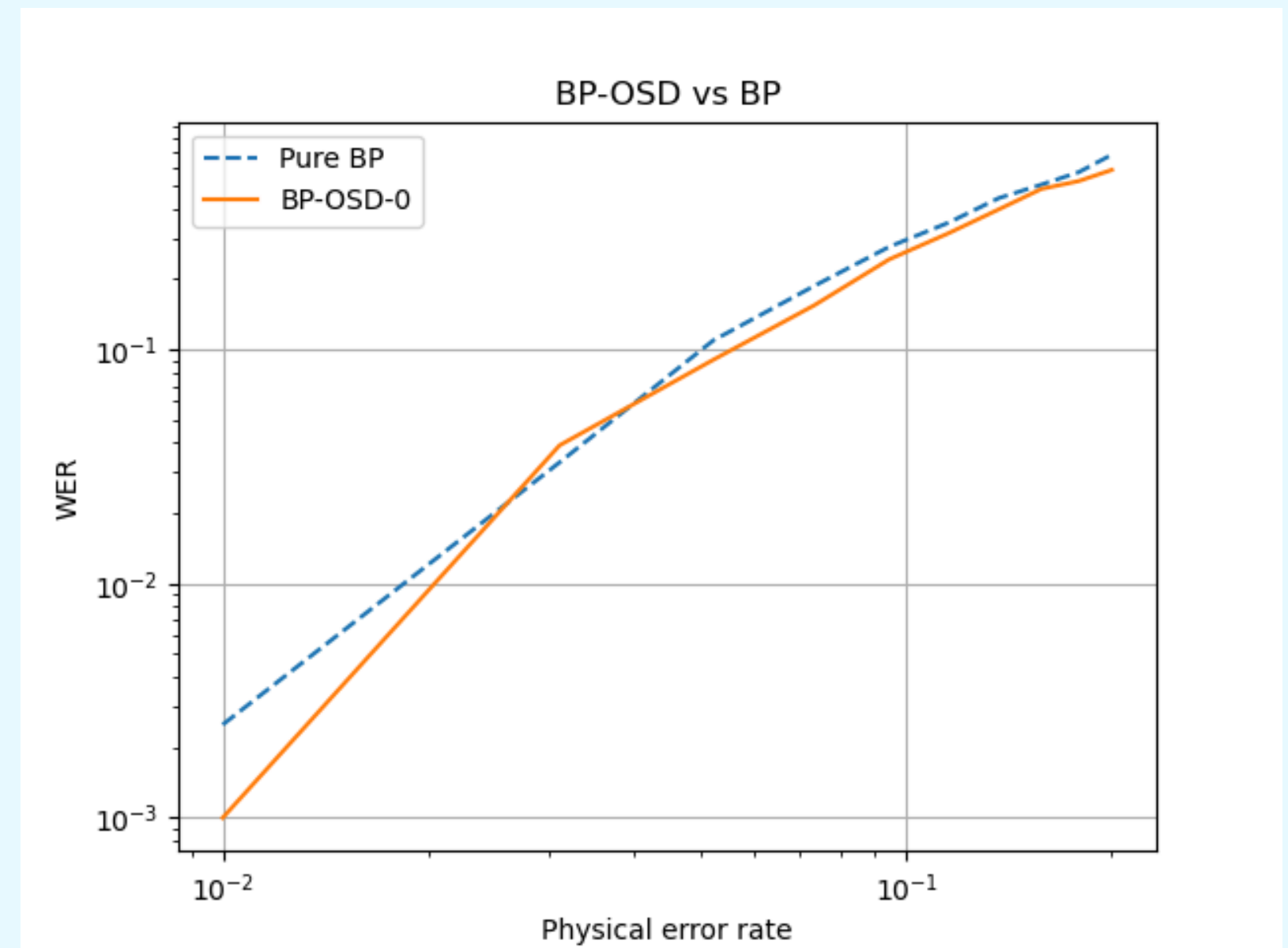
OSD post-processing resolves residual errors BP leaves unsatisfied

- **Error Curves -**

BP + OSD shows significant low logical error rate vs BP alone

- **Key insight -**

Syndrome-only BP decoding combined with reliability-sorted OSD correction yields a robust quantum decoder.



THE FULL DECODING PIPELINE

- The Quantum-Classical Bridge: We successfully map fragile quantum states and operators into robust classical data using the binary symplectic representation.
- Safe Extraction: Through carefully designed stabilizer circuits, we measure the classical syndrome (s) without destroying the logical quantum information.
- The Decoder's Job: This syndrome vector is the exact input required for our Belief Propagation (BP) and OSD algorithms to estimate the most likely physical error (e).
- The Ultimate Goal: By bridging quantum hardware with efficient classical algorithms, Quantum LDPC codes pave the way for highly scalable, fault-tolerant quantum computing.

The background features several abstract geometric shapes in two shades of blue. In the top right, there is a small dark blue circle and a larger light blue semi-circle. On the right side, there is a light blue semi-circle and a dark blue semi-circle. In the bottom right, there is a dark blue circle. On the left side, there is a dark blue circle, a light blue semi-circle, a dark blue semi-circle, and a light blue semi-circle. At the bottom left, there is a dark blue semi-circle and a light blue semi-circle.

**THANK
YOU**